Tutorial 6 - Digital Systems

1. A set-dominate latch has a set and reset input. It differs from a conventional SR latch in that an attempt to simultaneously set and reset results in setting the latch. The circuit for this is shown in Figure 1. Obtain the truth table (next-state table) for this latch.

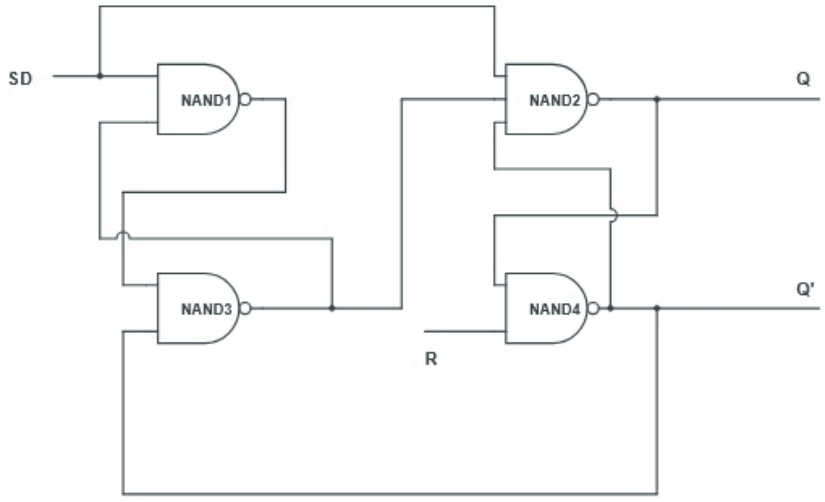


Figure 1: Set-dominate Latch

2. Consider the T flip-flop in Figure 2. Choose appropriate waveforms for the Clock signal and the T input and draw a timing diagram. Show Q as well as $\overline{Q}$.

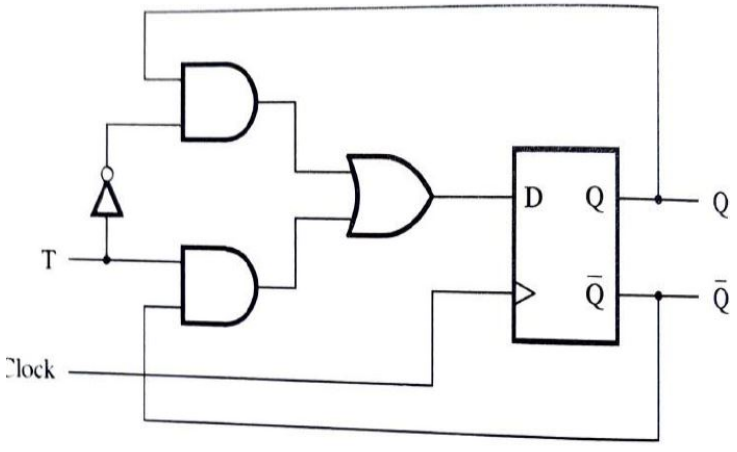


Figure 2: T flip-flop

3. Consider the JK flip-flop shown in Figure 3. Write the characteristic equation and the characteristic table for this circuit.

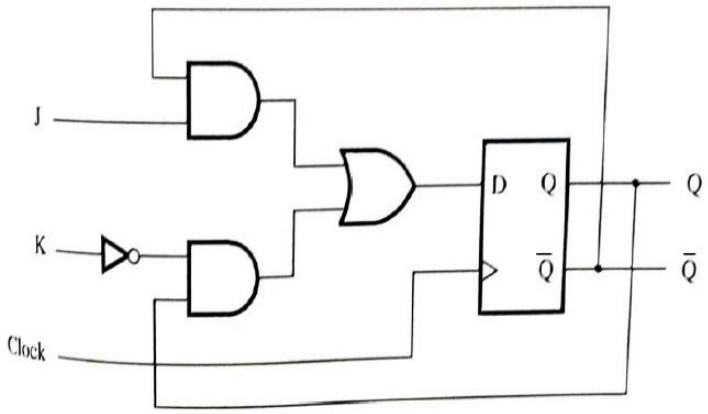


Figure 3: JK Flip-Flop

4. A four-bit shift register, which shifts data right ($Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow Q_3$) is shown in Figure 4. It is provided an initial state of $Q_0Q_1Q_2Q_3 = 1000$.

(a) Write the 4-bit values for Q_0, Q_1, Q_2, and Q_3 after each clock pulse until the pattern 1000 reappears.

(b) To what values should the shift register be initialized so that the pattern 1000 occurs after 2 clock pulses?

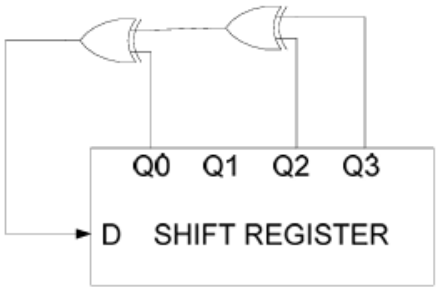


Figure 4: 4-bit Shift Register

5. Construct the truth table (next state table) of the sequential circuit in Figure 5 and comment on its operation.

6. Trace the outputs of a D latch and D flip-flop (positive-edge-triggered) for the input and clock signals provided in Figure 6. Compare the outputs and comment on the observations.

7. The initial content of a four-bit register is 0110. The register is shifted six times to the right with the serial input of 1011100. Assume the rightmost bit is the LSB. What is the content of the register after each shift ?

8. A 74178 shift register is described by Figure 7. All state changes occur on the 1-0 transition of the clock. The shift register is connected as shown in Figure 7. Complete the timing diagram.

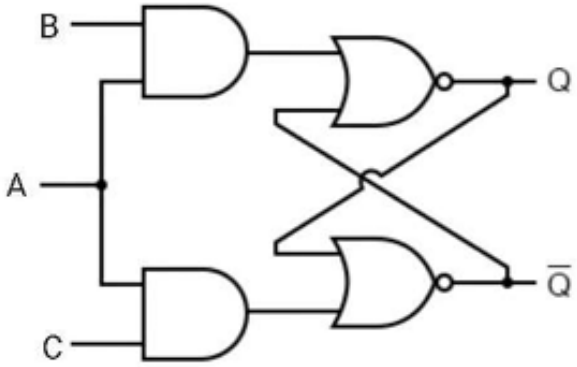


Figure 5: Sequential Circuit

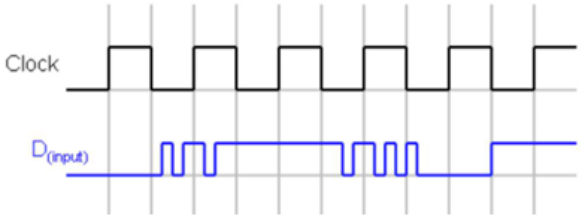
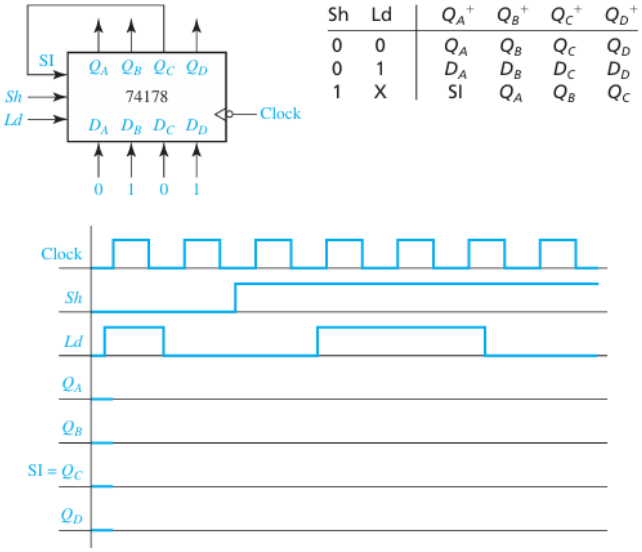


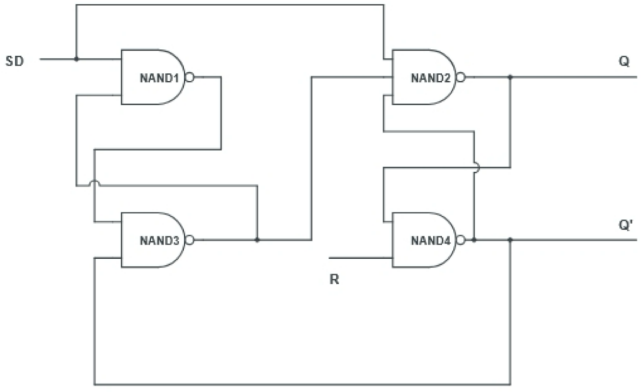
Figure 6: Timing Diagram



Sh	Ld	Q_A^+	Q_B^+	Q_C^+	Q_D^+
0	0	Q_A	Q_B	Q_C	Q_D
0	1	D_A	D_B	D_C	D_D
1	X	SI	Q_A	Q_B	Q_C

Figure 7: 74178 Shift register

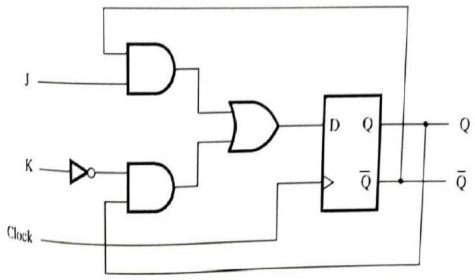
1. A set-dominate latch has a set and reset input. It differs from a conventional SR latch in that an attempt to simultaneously set and reset results in setting the latch. The circuit for this is shown in Figure 1. Obtain the truth table (next-state table) for this latch.



SD	R	Q(t)	Q(t+1)	
0	1	X	1	
1	1	1	1	→ memory
1	0	X	0	
1	1	0	0	→ memory
0	0	X	1	→ set dominant

3. Consider the JK flip-flop shown in Figure 3. Write the characteristic equation and the characteristic table for this circuit.

J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	Q'(t)



$D = JQ' + K'Q$

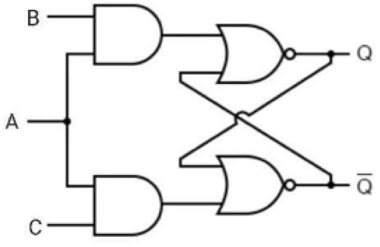
$Q(t+1) = J.\overline{Q(t+1)} + \overline{K}.Q(t)$

↳ characteristic equation

5. Construct the truth table (next state table) of the sequential circuit in Figure 5 and comment on its operation.

(C → S-R latch with enable input → A, B)

A	B	C	Q(t+1)
0	X	X	Q(t)
1	0	0	Q(t)
1	0	1	1
1	1	0	0
1	1	1	Invalid input



4. A four-bit shift register, which shifts data right ($Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow Q_3$) is shown in Figure 4. It is provided an initial state of $Q_0Q_1Q_2Q_3 = 1000$.

(a) Write the 4-bit values for Q_0, Q_1, Q_2, and Q_3 after each clock pulse until the pattern 1000 reappears.

(b) To what values should the shift register be initialized so that the pattern 1000 occurs after 2 clock pulses?

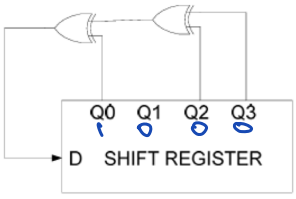


Figure 4: 4-bit Shift Register

$D = Q_0 \oplus (Q_2 \oplus Q_3)$

$= Q_0 \oplus (Q_2\bar{Q}_3 + \bar{Q}_2Q_3)$

$= Q_0(Q_2Q_3 + \bar{Q}_2\bar{Q}_3)$

$+ \bar{Q}_0(Q_2\bar{Q}_3 + \bar{Q}_2Q_3)$

odd number of 1's = 1

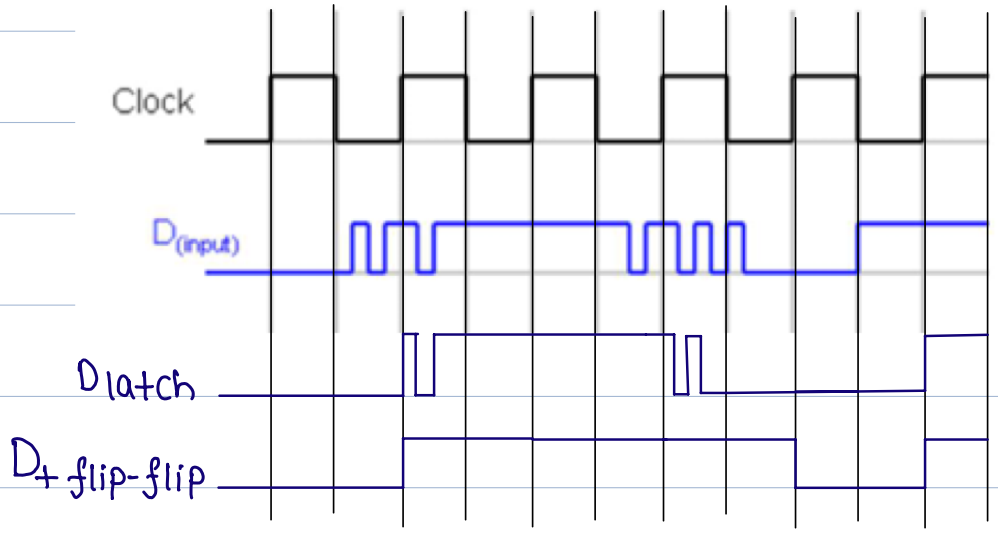
(a)

Q_0	Q_1	Q_2	Q_3	D
1	0	0	0	1
1	1	0	0	1
1	1	1	0	0
0	1	1	1	0
0	0	1	1	0
0	0	0	1	1
1	0	0	0	

6 cycles were required to get 1000 again

(b) If we initialized $Q_0Q_1Q_2Q_3$ as 0011, then after 2 clock pulses we get 1000

6. Trace the outputs of a D latch and D flip-flop (positive-edge-triggered) for the input and clock signals provided in Figure 6. Compare the outputs and comment on the observations.



The flip-flop output is more stable as it triggers only in the positive edge hence reducing glitches

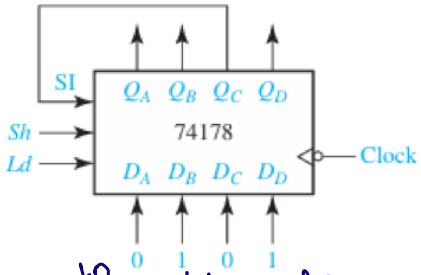
7. The initial content of a four-bit register is 0110. The register is shifted six times to the right with the serial input of 1011100. Assume the rightmost bit is the LSB. What is the content of the register after each shift ?

Shift No.	In	Q_0	Q_1	Q_2	Q_3
	0	0	1	1	0
1)	0	0	0	1	1
2)	1	0	0	0	1
3)	1	1	0	0	0
4)	1	1	1	0	0
5)	0	1	1	1	0

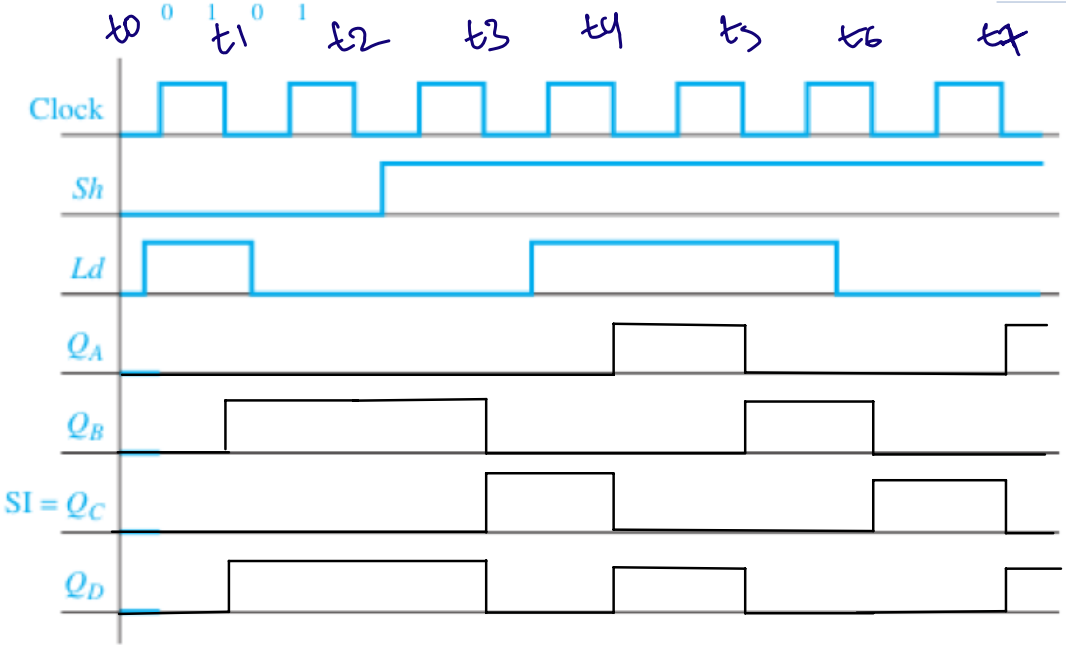
6)	1	0	1	1	1
7)		1	0	1	1

8. A 74178 shift register is described by Figure 7. All state changes occur on the 1-0 transition of the clock. The shift register is connected as shown in Figure 7. Complete the timing diagram.

Sh → Shift
Ld → load



Sh	Ld	Q_A^+	Q_B^+	Q_C^+	Q_D^+
0	0	Q_A	Q_B	Q_C	Q_D
0	1	D_A	D_B	D_C	D_D
1	X	SI	Q_A	Q_B	Q_C



t	Q_A	Q_B	Q_C	Q_D
t0	0	0	0	0
t1	0	1	0	1

t2	0	1	0	1
t3	0	0	1	0
t4	1	0	0	1
t5	0	1	0	0
t6	0	0	1	0
t7	1	0	0	1

2. Consider the T flip-flop in Figure 2. Choose appropriate waveforms for the Clock signal and the T input and draw a timing diagram. Show Q as well as $\overline{Q}$.

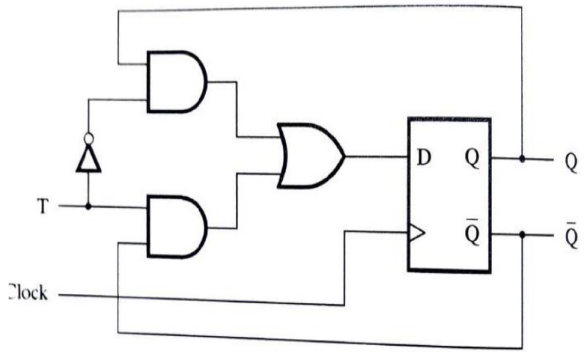


Figure 2: T flip-flop

